

Skills Summary

Square and Cube Roots

Use this reference while completing your worksheet or when studying.

Skill 1 — Evaluating perfect square roots

Rule: $\sqrt{N} = s$ exactly when $s \times s = N$ (and s is not negative).

Perfect squares worth knowing by sight:

1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225

Example: Evaluate $\sqrt{196}$.

Step 1. The radicand is a perfect square, so I hunt for the factor that repeats twice — no estimating needed.

Step 2. $10 \times 10 = 100$ is too small and $20 \times 20 = 400$ is too big, so try the teens: $14 \times 14 = 196$.
 $\Rightarrow \sqrt{196} = 14$

Try it: Evaluate $\sqrt{121}$.

check: 11

Skill 2 — Evaluating perfect cube roots

Rule: $\sqrt[3]{N} = s$ exactly when $s \times s \times s = N$.

Perfect cubes worth knowing by sight: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000.

Example: Evaluate $\sqrt[3]{343}$.

Step 1. A cube root asks for the factor used three times, so I test cubes rather than squares.

Step 2. $6 \times 6 \times 6 = 216$ is too small; $7 \times 7 \times 7 = 343$ lands exactly.
 $\Rightarrow \sqrt[3]{343} = 7$

Try it: Evaluate $\sqrt[3]{1000}$.

check: 10

(!) Watch out: $\sqrt{64} = 8$ but $\sqrt[3]{64} = 4$. The same radicand has a different square root and cube root — always read the little index first.

Skill 3 — Roots inside a larger expression

Rule: a root is a grouping symbol. Evaluate every root first, then do the multiplying, then the adding and subtracting. A number written in front of a root multiplies the *value* of the root — it never moves inside.

Example: Evaluate $2\sqrt{36} + \sqrt[3]{27}$.

Step 1. Two roots and a coefficient: order of operations says roots first, then the multiplication, then the sum.

Step 2. $\sqrt{36} = 6$ and $2 \times 6 = 12$; $\sqrt[3]{27} = 3$; $12 + 3 = 15$.

$$\Rightarrow 2\sqrt{36} + \sqrt[3]{27} = \mathbf{15}$$

Try it: Evaluate $\sqrt{400} - 3\sqrt[3]{8}$.

check: 14